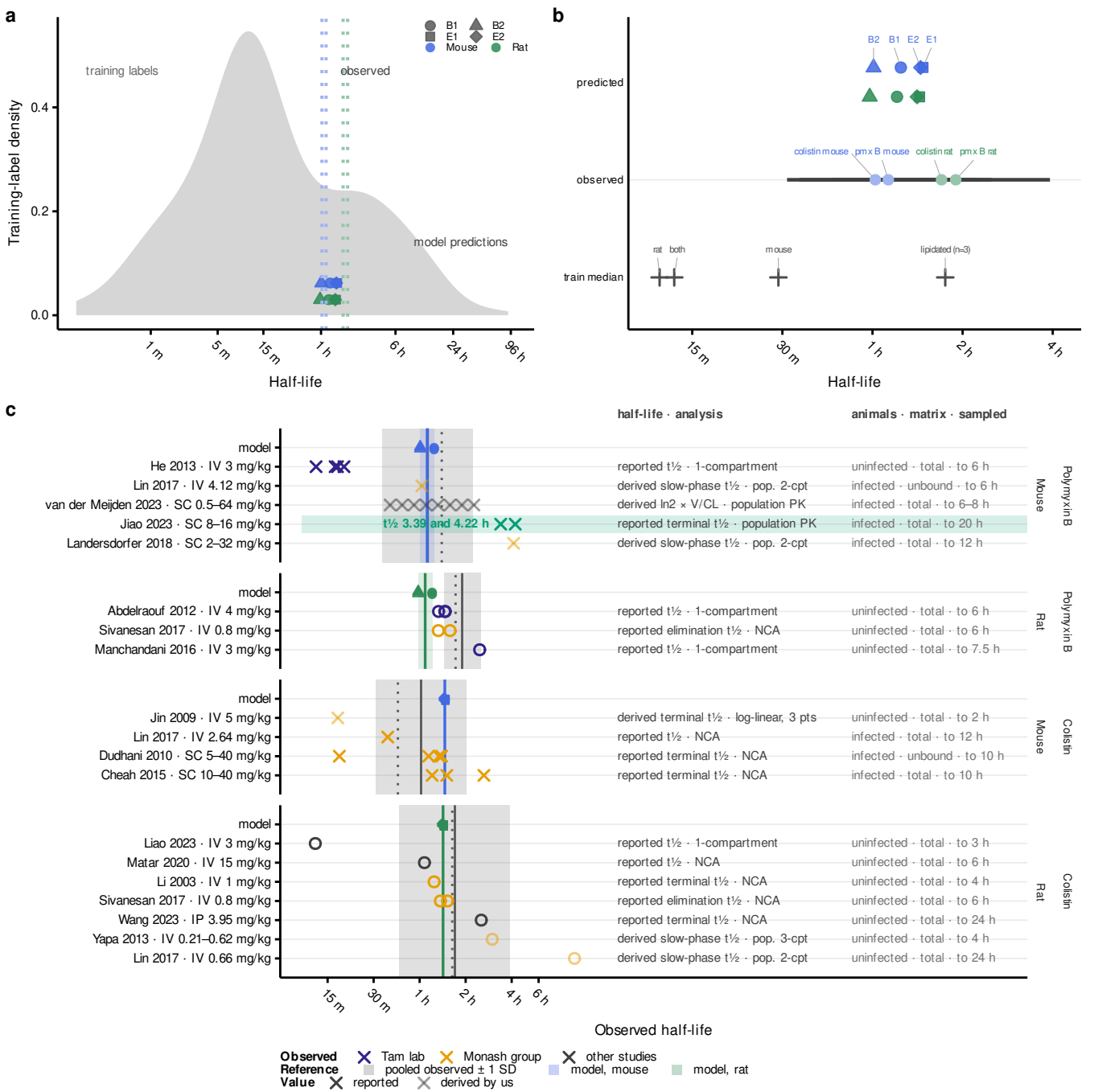
a
b
c
Training-label density
0.0
0.2
0.4
training labels
observed
model predictions
B1
B2
E1
E2
Mouse
Rat
1 m
5 m
15 m
1 h
6 h
24 h
96 h
Half-life
predicted
observed
train median
B2
B1
E2
E1
colistin mouse
pmx B mouse
colistin rat
pmx B rat
rat
both
mouse
lipidated (n=3)
15 m
30 m
1 h
2 h
4 h
Half-life
half-life · analysis
animals · matrix · sampled
model
He 2013 · IV 3 mg/kg
Lin 2017 · IV 4.12 mg/kg
van der Meijden 2023 · SC 0.5–64 mg/kg
Jiao 2023 · SC 8–16 mg/kg
Landersdorfer 2018 · SC 2–32 mg/kg
t½ 3.39 and 4.22 h
reported t½ · 1-compartment
derived slow-phase t½ · pop. 2-cpt
derived ln2 × V/CL · population PK
reported terminal t½ · population PK
derived slow-phase t½ · pop. 2-cpt
uninfected · total · to 6 h
infected · unbound · to 6 h
infected · total · to 6–8 h
infected · total · to 20 h
infected · total · to 12 h
Polymyxin B
Mouse
Abdelraouf 2012 · IV 4 mg/kg
Sivanesan 2017 · IV 0.8 mg/kg
Manchandani 2016 · IV 3 mg/kg
reported t½ · 1-compartment
reported elimination t½ · NCA
reported t½ · 1-compartment
uninfected · total · to 6 h
uninfected · total · to 6 h
uninfected · total · to 7.5 h
Polymyxin B
Rat
Jin 2009 · IV 5 mg/kg
Lin 2017 · IV 2.64 mg/kg
Dudhani 2010 · SC 5–40 mg/kg
Cheah 2015 · SC 10–40 mg/kg
derived terminal t½ · log-linear, 3 pts
reported t½ · NCA
reported terminal t½ · NCA
reported terminal t½ · NCA
uninfected · total · to 2 h
infected · total · to 12 h
infected · unbound · to 10 h
infected · total · to 10 h
Colistin
Mouse
Liao 2023 · IV 3 mg/kg
Matar 2020 · IV 15 mg/kg
Li 2003 · IV 1 mg/kg
Sivanesan 2017 · IV 0.8 mg/kg
Wang 2023 · IP 3.95 mg/kg
Yapa 2013 · IV 0.21–0.62 mg/kg
Lin 2017 · IV 0.66 mg/kg
reported t½ · 1-compartment
reported t½ · NCA
reported terminal t½ · NCA
reported elimination t½ · NCA
reported terminal t½ · NCA
derived slow-phase t½ · pop. 3-cpt
derived slow-phase t½ · pop. 2-cpt
uninfected · total · to 3 h
uninfected · total · to 6 h
uninfected · total · to 4 h
uninfected · total · to 6 h
uninfected · total · to 24 h
uninfected · total · to 4 h
uninfected · total · to 24 h
Colistin
Rat
15 m
30 m
1 h
2 h
4 h
6 h
Observed half-life
Observed
Tam lab
Monash group
other studies
Reference
pooled observed ± 1 SD
model, mouse
model, rat
Value
reported
derived by us